

#	S.No.	Problem	Challenge	Objective
1		Affordable Diagnostics	Developing a low-cost diagnostic tool that is accurate, easy to use, and accessible for resource-limited settings across the world.	Create an innovative diagnostic solution that provides early detection of diseases such as anemia, diabetes, or tuberculosis, ensuring scalability and affordability for underserved communities.
2		Personalized Medicine	Making personalized medicine accessible to diverse populations, especially in resource-constrained regions.	Develop an accessible framework to use genetic or biomarker data to tailor treatment plans, focusing on affordability, scalability, and equitable access to these technologies.
3		Plastic Waste Degradation	Addressing the environmental impact of plastic waste, especially in urban and marine ecosystems.	Create a bio-inspired system using microbes, enzymes, or plants to break down plastic waste efficiently and sustainably, reducing long-term pollution.
4		Biological Water Filtration	Ensuring access to clean drinking water in areas affected by contamination or water scarcity.	Design a bio-based filtration system that can remove pathogens, heavy metals, and other contaminants, ensuring affordability and ease of implementation in low-resource settings.
5		Sustainable Pest Control	Reducing reliance on chemical pesticides while maintaining crop productivity.	Develop a biological pest control system using engineered organisms or natural predators to create eco-friendly and sustainable pest management solutions.
6		Crop Management	Enhancing crop yield while addressing challenges like drought, nutrient deficiency, and pest attacks.	Propose a bio-based crop management system that combines soil health monitoring, pest control, and nutrient optimization for sustainable agriculture.
7		Soil Improvement	Restoring degraded soils and maintaining soil fertility in agricultural & forest lands.	Design a solution leveraging biofertilizers, microbial consortia, or plant-based systems to improve soil health of wild ecosystems and boost agricultural productivity sustainably.
8		Ethical Cattle Systems	Balancing the growing demand for livestock products with ethical and sustainable practices.	Create a system or framework that promotes ethical cattle farming practices, ensuring animal welfare, environmental sustainability, and economic feasibility.
9		Biology Education Toolkit	Teaching complex biology concepts effectively to high school students in underserved communities.	Develop an interactive educational toolkit that uses engaging activities, experiments, or visual aids to simplify topics like genetics, ecosystems, or synthetic biology.
10		Wildlife Monitoring	Tracking and conserving endangered species without disrupting their natural habitats.	Create a bio-inspired monitoring system using sensors, drones, or bioindicators to observe and protect wildlife in real time, ensuring minimal environmental impact.
11		Invasive Species Control	Managing invasive species that threaten native ecosystems and biodiversity.	Develop a sustainable, bio-based strategy to monitor and control invasive species, restoring balance to affected ecosystems while preserving native flora and fauna.
12		Food Technology and Packaging	Enhancing food quality, safety, and shelf life while addressing challenges like food spoilage, nutrient loss, and environmental impact.	Propose a bio-based food technology and packaging system that integrates precision food processing, functional food development, and precision fermentation to create sustainable, nutrient-rich, and long-lasting food products.